5A - 6A - 10A - Atmosphere - EvolLand

Test 3 Master Test Review

55 Review Questions

STUDENT EDITION

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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REVIEW

Test 3 Master Test Review

55 questions worth focusing on as you prepare for Test 3, organized by subject. Each one tells you where to find the answer. Work through them in your own words, typing your answers on the lines.

The Earth's Atmosphere

1. What is the atmosphere? Excluding water vapor, name the four main gases that make it up, with their percentages.

Find answers in: Module 5A, 5AE1

2. Name the four layers of the atmosphere in order, from the ground up, and the "pause" at the top of each. What is a pause?

Find answers in: Module 5A, 5AE1

3. In which two layers does the temperature DECREASE with increasing altitude, and in which two does it INCREASE? What heats each of the two warming layers?

Find answers in: Module 5A, 5AE1

4. What are jet streams? In the Northern Hemisphere, where are the two jet streams found, and why do airliners like to ride them?

Find answers in: Module 5A, 5AE1

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5. In which layer is the ozone layer, and what does it do? Which two gases in that layer protect life from deadly UV radiation? Why is ozone “good up high, bad nearby”?

Find answers in: Module 5A, 5AE1

6. What is the ionosphere, and where is it? Up to what frequency does it reflect radio waves?

Find answers in: Module 5A, 5AE1

7. The thermosphere can reach 2000° C. Why wouldn’t an astronaut out there feel hot? What is the difference between temperature and heat?

Find answers in: Module 5A, 5AE1, 5AE2

8. What causes the aurorae, and in which layer of the atmosphere do they occur?

Find answers in: Module 5A, 5AE1

9. What “sphere” harmlessly traps the charged particles of the solar wind?

Find answers in: Module 5A, 5AE1

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10. What is Earth-based weather? Name the six main ingredients of Earth-based weather.

Find answers in: Module 5A, 5AE2

11. Name four forms of precipitation, and describe each.

Find answers in: Module 5A, 5AE2

12. Are fog, dew, and clouds forms of precipitation? Explain.

Find answers in: Module 5A, 5AE2

13. What is lightning? You count 10 seconds between a lightning flash and its thunder. About how far away was the strike?

Find answers in: Module 5A, 5AE2

14. In summer, what are cumulonimbus (storm) clouds primarily composed of?

Find answers in: Module 5A, 5AE2

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15. What is a greenhouse gas? Name three. Why are nitrogen and oxygen NOT greenhouse gases?

Find answers in: Module 5A, 5AE2

16. What makes high and low pressure, and what weather does each bring?

Find answers in: Module 5A, 5AE2 (Homework #11)

17. In the Northern Hemisphere, which way does air circle a high, and which way does it circle a low?

Find answers in: Module 5A, 5AE2 (Homework #11)

Atmospheric Pressure Exercise

18. Define pressure and atmospheric pressure. How does atmospheric pressure change as you go higher, and why?

Find answers in: Atmospheric Pressure exercise

19. Who invented the mercury barometer? How tall is the mercury column at sea level, and what holds it up?

Find answers in: Atmospheric Pressure exercise

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20. Give the value of one Atmosphere in millibars, psi, and inches of mercury (in Hg).

Find answers in: Atmospheric Pressure exercise

Ancient Atmospheres

21. Once early Earth cooled below the boiling point of water, what happened, for about how long, and what was the result?

Find answers in: Ancient Atmospheres video (Module 5A, Homework #6)

22. What was the Great Oxygenation Event, and which organisms caused it?

Find answers in: Ancient Atmospheres video (Module 5A, Homework #6)

EvolLand Exercise

23. About how many millions of years ago was the Cambrian Explosion, and what happened in it?

Find answers in: EvolLand exercise

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24. As oxygen in the atmosphere rose, life moved from the sea to the land. About how high did oxygen peak, and when? What happened to flying insects?

Find answers in: EvolLand exercise (Module 5A, Homework #6)

25. In years, about how long ago did our species, Homo sapiens, evolve?

Find answers in: EvolLand exercise

The Nature of Light

26. What is optics?

Find answers in: Module 6A, 6AE1

27. What is the simple definition of light?

Find answers in: Module 6A, 6AE1

28. Light is described as a dual transverse, non-mechanical, self-propagating wave. What do “dual transverse,” “self-propagating,” and “non-mechanical” mean?

Find answers in: Module 6A, 6AE1

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29. Rounded, what is the speed of light in a vacuum, in km/s and in miles/s?

Find answers in: Module 6A, 6AE1

30. Who first used astronomy to determine the speed of light, when, and how?

Find answers in: Module 6A

31. Is light a wave or a particle? Why is light's nature called contradictory?

Find answers in: Module 6A, 6AE1

32. What are the two schools of light? For each, give its founder, what he said light is made of, and how the school draws light's path.

Find answers in: Module 6A, 6AE1

33. What do "temporally coherent" and "spatially coherent" mean? Why is laser light coherent, and why does it dim so slowly with distance?

Find answers in: Module 6A, 6AE1

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34. List the electromagnetic spectrum in order of increasing frequency. About what range of wavelengths can humans see? Are cosmic rays part of the spectrum?

Find answers in: Module 6A, 6AE1

35. What are the three ways light interacts with matter? Define each.

Find answers in: Module 6A, 6AE1

36. Name and define the five classical phenomena of light. Which model of light explains all five, and which one can the particle model adequately explain?

Find answers in: Module 6A, 6AE1

37. What is the difference between specular and diffuse reflection? Do all shiny objects exhibit specular reflection?

Find answers in: Module 6A

38. What causes refraction? When light passes from air into water, does it speed up or slow down, and does the ray bend toward or away from the normal?

Find answers in: Module 6A

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39. True or false: Cherenkov radiation is the blue glow given off when charged particles, such as electrons, travel through water faster than light itself travels in water.

Find answers in: Module 6A

The Exploration of Space

40. What was Sputnik 1? Who launched it? Why could no one doubt it was real?

Find answers in: Module 10A

41. What was the first satellite the United States successfully launched? What did it discover?

Find answers in: Module 10A

42. What made Telstar 1 historic?

Find answers in: Module 10A

43. Who was the first person to orbit the Earth?

Find answers in: Module 10A

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44. Who were the first and second Americans in space?

Find answers in: Module 10A

45. Who was the first American to orbit the Earth?

Find answers in: Module 10A

46. What were the two parts of President Kennedy's 1961 challenge? Which Apollo 11 astronauts first stepped on the Moon, when, and where did they land?

Find answers in: Module 10A

47. How many Apollo missions set out to land on the Moon, how many succeeded, and how many men have walked on the Moon?

Find answers in: Module 10A

48. What famous "first" in aviation belongs to Orville and Wilbur Wright? When and where did it happen?

Find answers in: Module 10A

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49. What famous “first” in aviation belongs to Bessie Coleman?

Find answers in: Module 10A

50. What famous “first” in aviation belongs to Charles Lindbergh? In what plane, and from where to where?

Find answers in: Module 10A

51. What famous “first” in aviation belongs to Amelia Earhart?

Find answers in: Module 10A

52. Which uncrewed mission produced the first close-up images of Jupiter? What else was first about it?

Find answers in: Module 10A

53. Which uncrewed mission produced the first close-up images of Saturn?

Find answers in: Module 10A

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54. What were the Viking missions, and what was their key first?

Find answers in: Module 10A

55. Which spacecraft is the only one to have visited Uranus and Neptune?

Find answers in: Module 10A

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